

Slip 1

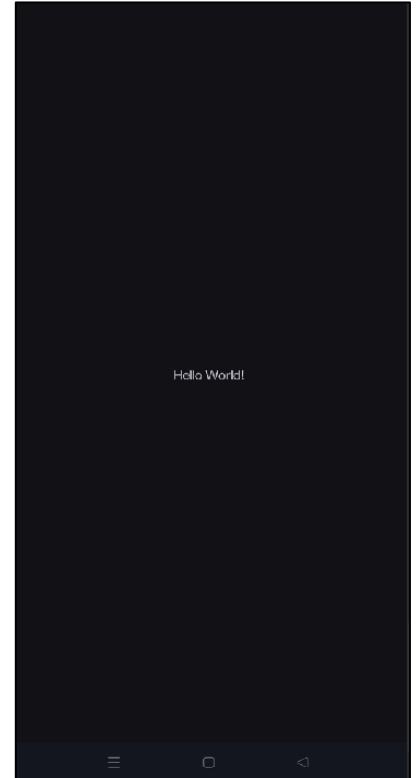
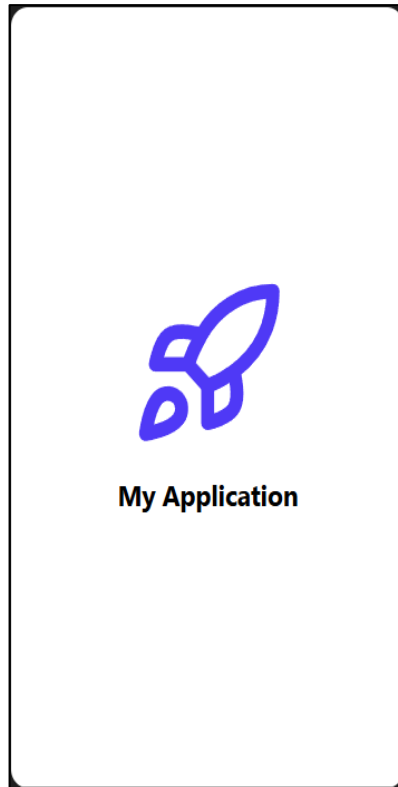
Q.1) Write an application to create a splash screen.

activity_splash.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFFFFF"
    android:gravity="center"
    android:orientation="vertical">

    <ImageView
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:src="@drawable/ic_launcher" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Application"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp" />
</LinearLayout>
```



Splash_activity.java

```
package com.example.splash_screen;

import android.content.Intent;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Splash_activity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_splash);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        Thread t = new Thread(){
            @Override
            public void run() {
                try {
                    sleep(3000);
                } catch (Exception e) {
                    throw new RuntimeException(e);
                }
            }
        };
        t.start();
    }
}
```

```

        } finally {
            startActivity(new Intent(Splash_activity.this, MainActivity.class));
        }
    }
};
t.start();
}
}
}

```

AndroidManifest.xml

```

<activity android:name=".Splash_activity" android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
<activity android:name=".MainActivity" />

```

Q.2) Create table Student (roll_no, name, address, percentage). Create Application for performing the following operation on the table...

DBHelper.java

```

package com.example.studentdb;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "Student.db";
    public static final int DB_VERSION = 1;
    public static final String TABLE_NAME = "Student";

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String query = "CREATE TABLE " + TABLE_NAME + " (" +
            "roll_no INTEGER PRIMARY KEY, " +
            "name TEXT, " +
            "address TEXT, " +
            "percentage REAL)";
        db.execSQL(query);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public void insertStudent(int roll, String name, String address, double percentage) {

```

```

        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("roll_no", roll);
        cv.put("name", name);
        cv.put("address", address);
        cv.put("percentage", percentage);
        db.insert(TABLE_NAME, null, cv);
    }

    public void insertFiveStudents() {
        insertStudent(1, "Amit", "Pune", 85.5);
        insertStudent(2, "Neha", "Mumbai", 78.0);
        insertStudent(3, "Rahul", "Nashik", 82.3);
        insertStudent(4, "Sneha", "Nagpur", 90.2);
        insertStudent(5, "Priya", "Aurangabad", 88.8);
    }

    public Cursor getAllStudents() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:padding="16dp"
            android:textSize="18sp"
            android:textColor="#000000"
            android:lineSpacingExtra="4dp"/>

    </ScrollView>
</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java

```

package com.example.studentdb;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;

```



```

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    TextView textView;
    DBHelper dbHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        textView = findViewById(R.id.textView);
        dbHelper = new DBHelper(this);

        // Insert records
        dbHelper.insertFiveStudents();

        // Fetch data
        Cursor cursor = dbHelper.getAllStudents();
        StringBuilder data = new StringBuilder();

        if (cursor.moveToFirst()) {
            do {
                data.append("Roll No: ").append(cursor.getInt(0)).append("\n");
                data.append("Name: ").append(cursor.getString(1)).append("\n");
                data.append("Address: ").append(cursor.getString(2)).append("\n");
                data.append("Percentage: ").append(cursor.getDouble(3)).append("\n\n");
            } while (cursor.moveToNext());
        }
        cursor.close();
        textView.setText(data.toString());
    }
}

```

Slip 2

Q.1) Create an application that allows the user to enter a number in the textbox. Check whether the number in the textbox is perfect number or not.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"

```

```
android:padding="20dp">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Enter a Number"
    android:textSize="18sp"
    android:textStyle="bold" />
```

```
<EditText
    android:id="@+id/editTextText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="e.g. 6, 28, 496"
    android:inputType="number"
    android:textAlignment="center" />
```

```
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Check Perfect Number" />
```

```
<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="30dp"
    android:text="Result will appear here"
    android:textSize="20sp"
    android:textColor="#000000"
    android:textStyle="bold" />
```

```
</LinearLayout>
```

MainActivity.java

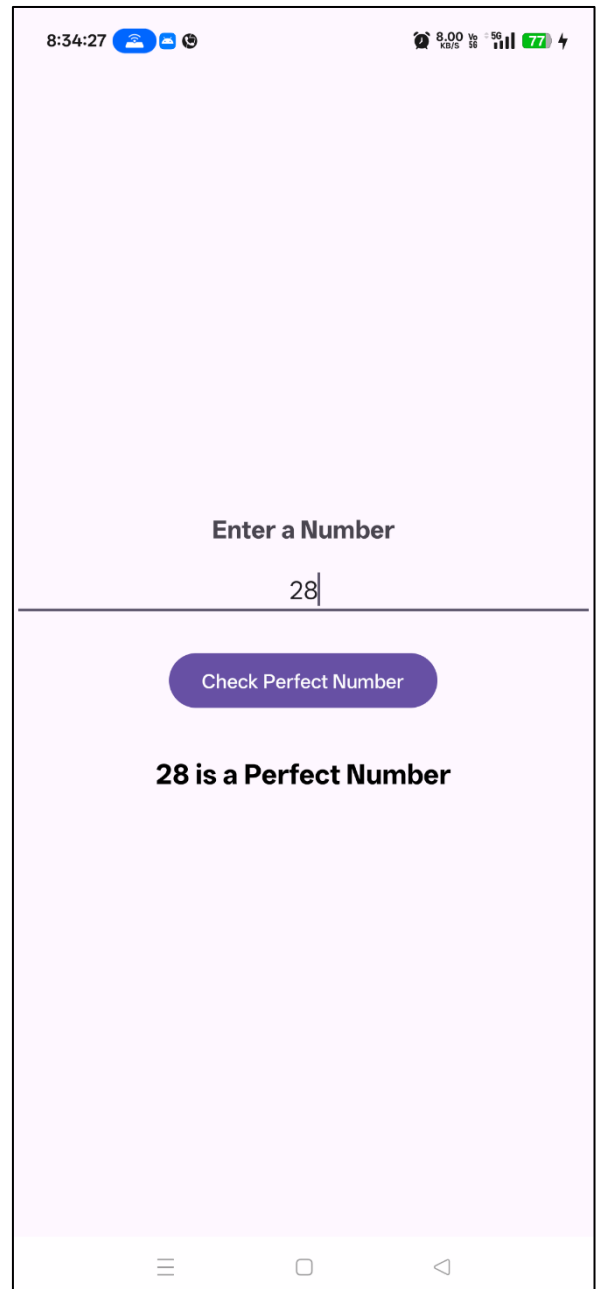
```
package com.example.print_number;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
    EditText e1;
    Button b1;
    TextView t1;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
```



```

setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});

e1 = findViewById(R.id.editTextText);
b1 = findViewById(R.id.button);
t1 = findViewById(R.id.textView2);

b1.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        String val = e1.getText().toString();

        if (!val.isEmpty()) {
            int num = Integer.parseInt(val);
            int sum = 0;

            // Logic to find divisors and sum them
            for (int i = 1; i < num; i++) {
                if (num % i == 0) {
                    sum += i;
                }
            }

            if (sum == num && num > 0) {
                t1.setText(num + " is a Perfect Number");
            } else {
                t1.setText(num + " is NOT a Perfect Number");
            }
        } else {
            t1.setText("Please enter a value");
        }
    }
});
}
}

```

Q.2) Java Android Program to perform all arithmetic Operations using Calculator.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/display"
        android:layout_width="match_parent"
        android:layout_height="60dp"
        android:textSize="24sp"
        android:hint="0"
        android:gravity="end"
        android:inputType="none"/>

```

```

<GridLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:columnCount="4"
    android:rowCount="5"
    android:layout_marginTop="20dp">

    <Button android:text="1" android:onClick="onDigit"/>
    <Button android:text="2" android:onClick="onDigit"/>
    <Button android:text="3" android:onClick="onDigit"/>
    <Button android:text="+" android:onClick="onOperator"/>

    <Button android:text="4" android:onClick="onDigit"/>
    <Button android:text="5" android:onClick="onDigit"/>
    <Button android:text="6" android:onClick="onDigit"/>
    <Button android:text="-" android:onClick="onOperator"/>

    <Button android:text="7" android:onClick="onDigit"/>
    <Button android:text="8" android:onClick="onDigit"/>
    <Button android:text="9" android:onClick="onDigit"/>
    <Button android:text="*" android:onClick="onOperator"/>

    <Button android:text="." android:onClick="onDigit"/>
    <Button android:text="0" android:onClick="onDigit"/>
    <Button android:text="C" android:onClick="onClear"/>
    <Button android:text="/" android:onClick="onOperator"/>

    <Button
        android:text="="
        android:layout_columnSpan="4"
        android:onClick="onEqual"/>
</GridLayout>

</LinearLayout>

```

MainActivity.java

```

package com.example.calculator_ui;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText display;

    double num1 = 0, num2 = 0;
    String operator = "";
    boolean isNewOperation = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

```

    display = findViewById(R.id.display);
}

// Digit button click
public void onDigit(View view) {
    Button btn = (Button) view;
    if (isNewOperation) {
        display.setText("");
        isNewOperation = false;
    }

    display.append(btn.getText().toString());

// Operator button click
public void onOperator(View view) {
    Button btn = (Button) view;
    num1 = Double.parseDouble(display.getText().toString());
    operator = btn.getText().toString();
    isNewOperation = true;
}

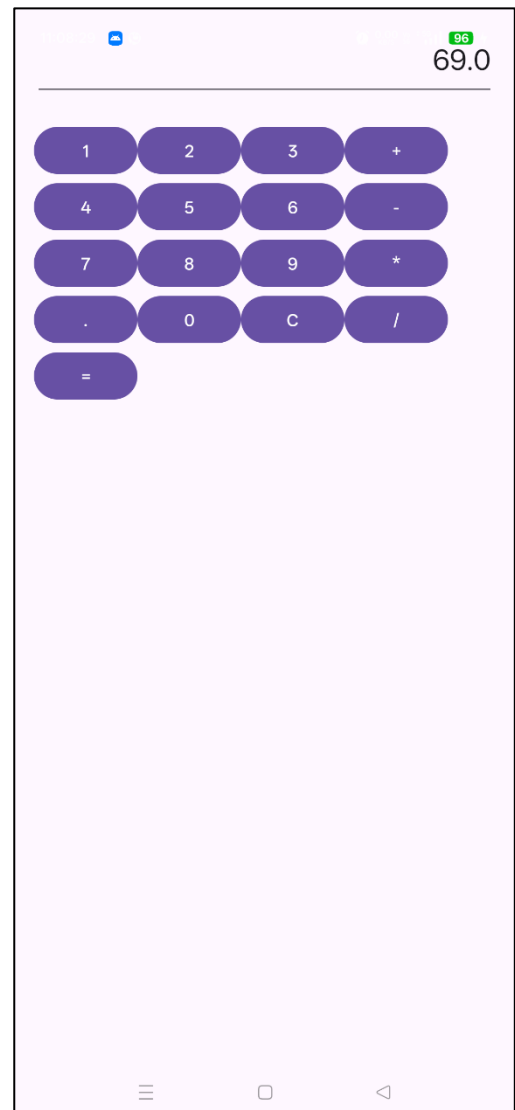
// Equal button click
public void onEqual(View view) {
    num2 = Double.parseDouble(display.getText().toString());
    double result = 0;

    switch (operator) {
        case "+":
            result = num1 + num2;
            break;
        case "-":
            result = num1 - num2;
            break;
        case "*":
            result = num1 * num2;
            break;
        case "/":
            if (num2 != 0)
                result = num1 / num2;
            else
                display.setText("Error");
            break;
    }

    display.setText(String.valueOf(result));
    isNewOperation = true;
}

// Clear button
public void onClear(View view) {
    display.setText("");
    num1 = num2 = 0;
    operator = "";
    isNewOperation = true;
}
}

```



Slip 3

Q.1) Create an application that allows the user to enter a number in the textbox. Check whether the number in the textbox is Armstrong or not.

activity_main.xml

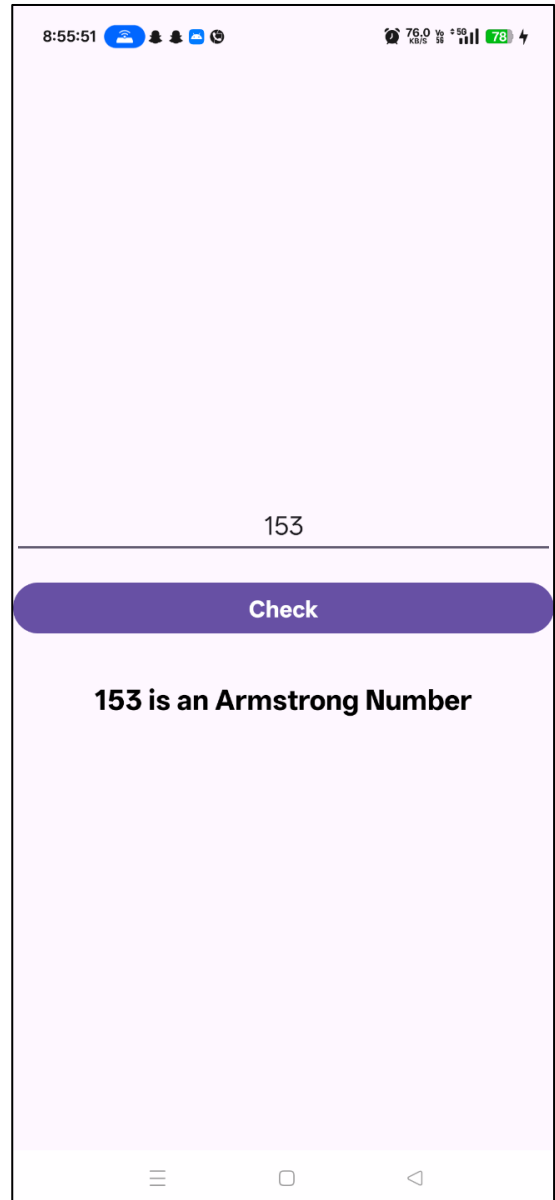
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

    <EditText
        android:id="@+id/editTextText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter a number (e.g. 153)"
        android:inputType="number"
        android:textAlignment="center"
        android:textSize="20sp"
        android:padding="12dp" />

    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Check"
        android:textSize="18sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text=""
        android:textColor="#000000"
        android:textSize="22sp"
        android:textStyle="bold" />

</LinearLayout>
```



MainActivity.java

```
package com.example.armstrong;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```

public class MainActivity extends AppCompatActivity {

    EditText e1;
    Button b1;
    TextView t1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        e1 = findViewById(R.id.editTextText);
        b1 = findViewById(R.id.button);
        t1 = findViewById(R.id.textView2);

        b1.setOnClickListener(view -> {
            String input = e1.getText().toString();

            if (input.isEmpty()) {
                t1.setText("Please enter a number");
                return;
            }

            int num = Integer.parseInt(input);
            int temp = num;
            int digits = 0;
            int result = 0;

            // Step 1: Count the number of digits
            while (temp != 0) {
                digits++;
                temp /= 10;
            }

            temp = num;

            // Step 2: Calculate Armstrong sum
            while (temp != 0) {
                int remainder = temp % 10;
                result += (int) Math.pow(remainder, digits);
                temp /= 10;
            }

            // Step 3: Display result
            if (result == num) {
                t1.setText(num + " is an Armstrong Number");
            } else {
                t1.setText(num + " is NOT an Armstrong Number");
            }
        });
    }
}

```

Q.2) Create an Android application which examine a phone number entered by a user with the given format. • Area code should be one of the following: 040, 041, 050, 0400, 044 • There should 6 - 8 numbers in telephone number (+ area code).

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/phoneInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone" />

    <Button
        android:id="@+id/validateBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Validate"
        android:layout_marginTop="10dp" />

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textSize="18sp"
        android:textStyle="bold" />

</LinearLayout>
```

MainActivity.java

```
package com.example.phone_validation;

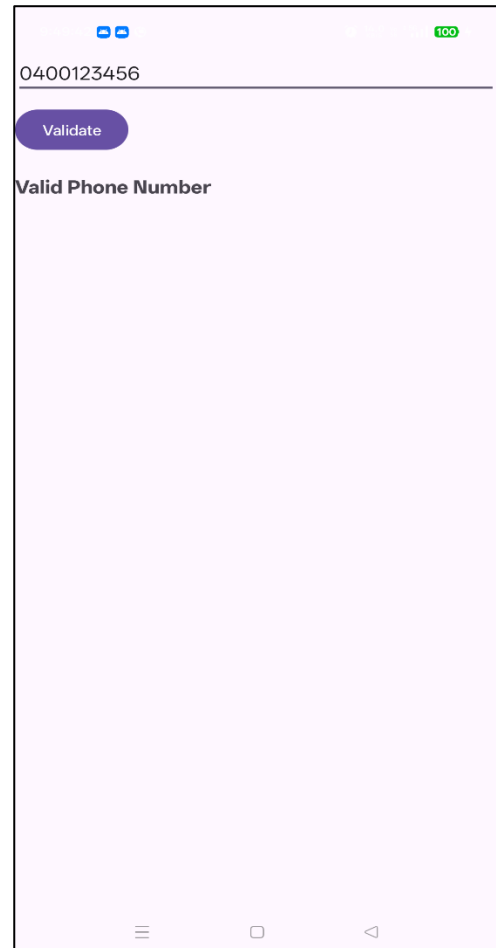
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.regex.Pattern;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```



```

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});

EditText phoneInput = findViewById(R.id.phoneInput);
Button validateBtn = findViewById(R.id.validateBtn);
TextView result = findViewById(R.id.result);

validateBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String phone = phoneInput.getText().toString();

        // The pattern matches numbers starting with 0400, 040, 041, 050, or 044
        // followed by 6 to 8 digits.
        String pattern = "^(0400|040|041|050|044)\\d{6,8}$";

        if (Pattern.matches(pattern, phone)) {
            result.setText("Valid Phone Number");
        } else {
            result.setText("Invalid Phone Number");
        }
    }
});
}
}

```

Slip 4

Q.1) Construct image switcher using setFactory().

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageSwitcher
        android:id="@+id/imageSwitcher"
        android:layout_width="300dp"
        android:layout_height="300dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.495"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.102" />

    <Button
        android:id="@+id/btnNext"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```

```

        android:layout_marginTop="20dp"
        android:text="Next Image"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.666" />

```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.image_switcher;
```

```

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageSwitcher;
import android.widget.ImageView;
import android.widget.ViewSwitcher;

```

```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```

```
public class MainActivity extends AppCompatActivity {
```

```

    ImageSwitcher imageSwitcher;
    Button btnNext;

```

```

    int[] images = {
        R.drawable.apple,
        R.drawable.group,
        R.drawable.panda
    };

```

```
int index = 0;
```

```
@Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

```

```

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });

```

```

    imageSwitcher = findViewById(R.id.imageSwitcher);
    btnNext = findViewById(R.id.btnNext);

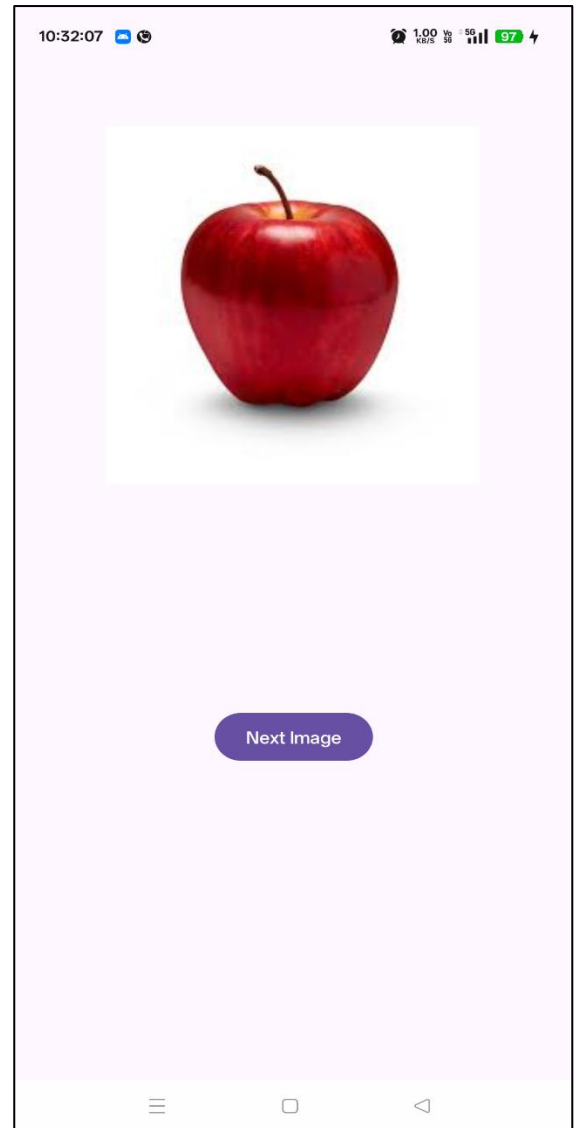
```

```
// setFactory() implementation
```

```

    imageSwitcher.setFactory(new ViewSwitcher.ViewFactory() {
        @Override
        public View makeView() {
            ImageView imageView = new ImageView(getApplicationContext());
            imageView.setScaleType(ImageView.ScaleType.FIT_CENTER);

```



```

        imageView.setLayoutParams(
            new ImageSwitcher.LayoutParams(
                ImageSwitcher.LayoutParams.MATCH_PARENT,
                ImageSwitcher.LayoutParams.MATCH_PARENT));
        return imageView;
    }
});

imageSwitcher.setImageResource(images[index]);

// Button click to change image
btnNext.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        index = (index + 1) % images.length;
        imageSwitcher.setImageResource(images[index]);
    }
});
}
}

```

Q.2) Write a program to search a specific location on Google Map.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center_horizontal">

    <EditText
        android:id="@+id/locationInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:hint="Enter Location (e.g., Pratibha College)"
        android:inputType="textPostalAddress" />

    <Button
        android:id="@+id/searchBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Search on Maps" />

</LinearLayout>

```

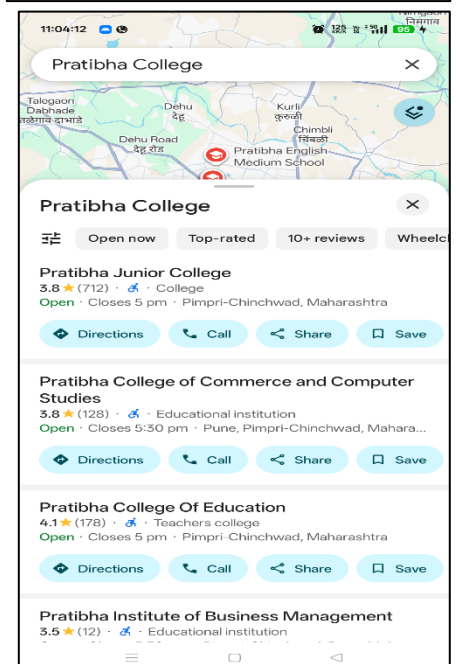
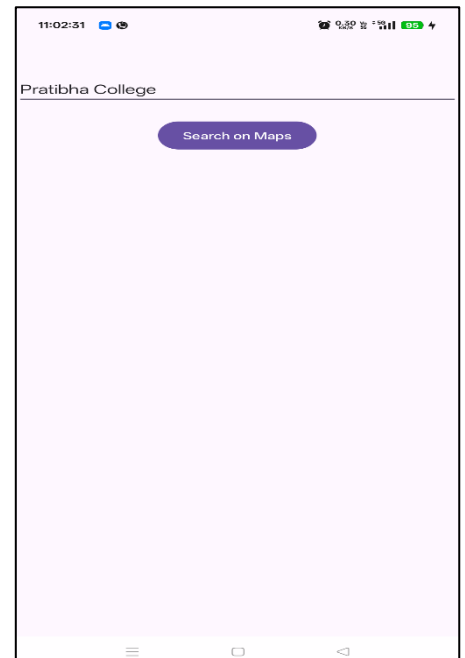
MainActivity.java

```

package com.example.gmap;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

```



```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Apply window insets for edge-to-edge support
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        EditText locationInput = findViewById(R.id.locationInput);
        Button searchBtn = findViewById(R.id.searchBtn);

        searchBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get text from input and launch Google Maps Intent
                String location = locationInput.getText().toString();
                Uri uri = Uri.parse("geo:0,0?q=" + location);
                Intent intent = new Intent(Intent.ACTION_VIEW, uri);
                startActivity(intent);
            }
        });
    }
}

```

Slip 5

Q.1) Java Android Program to Demonstrate Alert Dialog Box.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="16dp">

    <Button
        android:id="@+id/btnAlert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Alert Dialog" />

</LinearLayout>

```

MainActivity.java

```
package com.example.alert_box;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.DialogInterface;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;

public class MainActivity extends AppCompatActivity {

    Button btnAlert;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        btnAlert = findViewById(R.id.btnAlert);

        btnAlert.setOnClickListener(view -> {

            AlertDialog.Builder builder = new AlertDialog.Builder(MainActivity.this);

            builder.setTitle("Confirmation");
            builder.setMessage("Do you want to exit?");

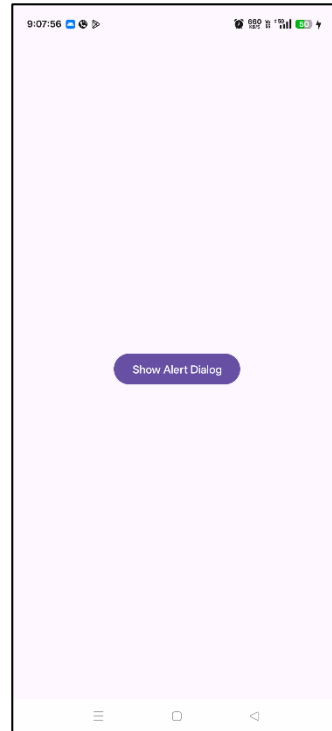
            builder.setPositiveButton("Yes", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked Yes", Toast.LENGTH_SHORT).show();

                finish(); // closes app
            });

            builder.setNegativeButton("No", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked No", Toast.LENGTH_SHORT).show();
            });

            builder.setNeutralButton("Cancel", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked Cancel", Toast.LENGTH_SHORT).show();
            });

            builder.show();
        });
    }
}
```



Q.2) Create an Android application which will ask the user to input his / her name. A message should display the two items concatenated in a label. Change the format of the label using radio buttons and check boxes for selection. The user can make the label text bold, underlined or italic as well as change its color. Also include buttons to display the message in the label, clear the text boxes as well as label. Finally exit.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <EditText
            android:id="@+id/etName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Your Name" />

        <TextView
            android:id="@+id/tvMessage"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="20sp"
            android:padding="10dp"
            android:text="Message will appear here" />

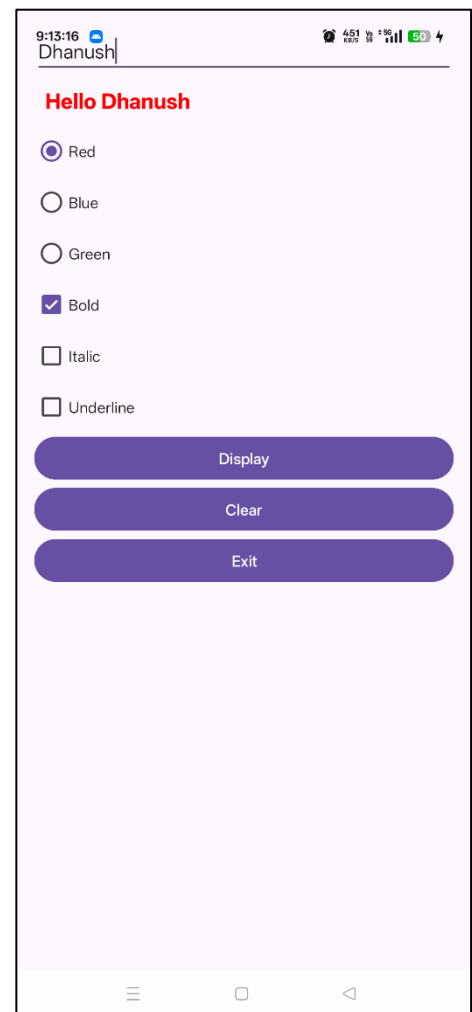
        <RadioGroup
            android:id="@+id/radioGroupColor"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">

            <RadioButton
                android:id="@+id/rbRed"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Red" />

            <RadioButton
                android:id="@+id/rbBlue"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Blue" />

            <RadioButton
                android:id="@+id/rbGreen"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Green" />
        </RadioGroup>

        <CheckBox
            android:id="@+id/cbBold"
            android:layout_width="wrap_content"
```



```
    android:layout_height="wrap_content"
    android:text="Bold" />
```

```
<CheckBox
    android:id="@+id/cbItalic"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Italic" />
```

```
<CheckBox
    android:id="@+id/cbUnderline"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Underline" />
```

```
<Button
    android:id="@+id/btnDisplay"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Display" />
```

```
<Button
    android:id="@+id/btnClear"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Clear" />
```

```
<Button
    android:id="@+id/btnExit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Exit" />
```

```
</LinearLayout>
```

```
</ScrollView>
```

MainActivity.java

```
package com.example.textformatter;
```

```
import android.os.Bundle;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.Typeface;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    EditText etName;
    TextView tvMessage;
    RadioGroup radioGroupColor;
```

```

RadioButton rbRed, rbBlue, rbGreen;
CheckBox cbBold, cbItalic, cbUnderline;
Button btnDisplay, btnClear, btnExit;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

    // Initialize Views
    etName = findViewById(R.id.etName);
    tvMessage = findViewById(R.id.tvMessage);
    radioGroupColor = findViewById(R.id.radioGroupColor);

    rbRed = findViewById(R.id.rbRed);
    rbBlue = findViewById(R.id.rbBlue);
    rbGreen = findViewById(R.id.rbGreen);

    cbBold = findViewById(R.id.cbBold);
    cbItalic = findViewById(R.id.cbItalic);
    cbUnderline = findViewById(R.id.cbUnderline);

    btnDisplay = findViewById(R.id.btnDisplay);
    btnClear = findViewById(R.id.btnClear);
    btnExit = findViewById(R.id.btnExit);

    // Display Button Logic
    btnDisplay.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String name = etName.getText().toString();
            String message = "Hello " + name;

            tvMessage.setText(message);

            // Color Change Logic
            if (rbRed.isChecked()) {
                tvMessage.setTextColor(Color.RED);
            } else if (rbBlue.isChecked()) {
                tvMessage.setTextColor(Color.BLUE);
            } else if (rbGreen.isChecked()) {
                tvMessage.setTextColor(Color.GREEN);
            }

            // Style Change Logic (Bold & Italic)
            int style = Typeface.NORMAL;

            if (cbBold.isChecked() && cbItalic.isChecked()) {
                style = Typeface.BOLD_ITALIC;
            } else if (cbBold.isChecked()) {
                style = Typeface.BOLD;
            } else if (cbItalic.isChecked()) {
                style = Typeface.ITALIC;
            }

            tvMessage.setTypeface(null, style);

            // Underline Logic
            if (cbUnderline.isChecked()) {

```

```

        tvMessage.setPaintFlags(tvMessage.getPaintFlags() | Paint.UNDERLINE_TEXT_FLAG);
    } else {
        tvMessage.setPaintFlags(tvMessage.getPaintFlags() & (~Paint.UNDERLINE_TEXT_FLAG));
    }
}
});

// Clear Button Logic
btnClear.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        etName.setText("");
        tvMessage.setText("Message will appear here");
        tvMessage.setTextColor(Color.BLACK); // Optional: Reset color to default
        tvMessage.setTypeface(null, Typeface.NORMAL); // Optional: Reset style to default
        tvMessage.setPaintFlags(tvMessage.getPaintFlags() & (~Paint.UNDERLINE_TEXT_FLAG)); // Optional:
Remove underline

        // Clear selections
        radioGroupColor.clearCheck();
        cbBold.setChecked(false);
        cbItalic.setChecked(false);
        cbUnderline.setChecked(false);
    }
});

// Exit Button Logic
btnExit.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        finish();
    }
});
}
}
}

```

Slip 6

Q.1) Java Android Program to demonstrate login form with validation.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="32dp">

    <TextView
        android:id="@+id/head"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="32dp"
        android:text="LOGIN FORM"
        android:textSize="24sp"
        android:textStyle="bold" />

```

```
<EditText
    android:id="@+id/editTextText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="16dp"
    android:hint="Username"
    android:inputType="text"
    android:minHeight="48dp" />
```

```
<EditText
    android:id="@+id/editTextText2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="32dp"
    android:hint="Password"
    android:inputType="textPassword"
    android:minHeight="48dp" />
```

```
<Button
    android:id="@+id/button"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:text="LOGIN"
    android:textStyle="bold" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.login_form;
```

```
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
```

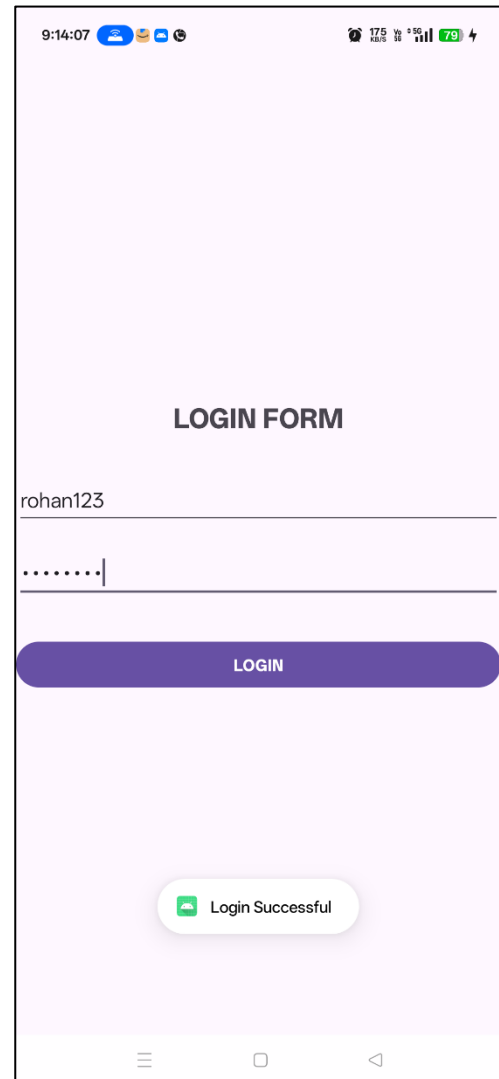
```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
```

```
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
```

```
        // Initialize UI components
```

```
        EditText usr = findViewById(R.id.editTextText);
        EditText pass = findViewById(R.id.editTextText2);
        Button b1 = findViewById(R.id.button);
```



```

// Set click listener
b1.setOnClickListener(view -> {
    String user = usr.getText().toString().trim();
    String password = pass.getText().toString().trim();

    // Validation logic
    if (user.isEmpty() || password.isEmpty()) {
        Toast.makeText(MainActivity.this, "Please enter both fields", Toast.LENGTH_SHORT).show();
    } else if (user.equals(password)) {
        // For this demo, login is successful if username equals password
        Toast.makeText(MainActivity.this, "Login Successful", Toast.LENGTH_LONG).show();
    } else {
        Toast.makeText(MainActivity.this, "Invalid Login", Toast.LENGTH_LONG).show();
    }
});
}
}

```

Q.2) Write a program to search a specific location on Google Map.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center_horizontal">

    <EditText
        android:id="@+id/locationInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:hint="Enter Location (e.g., Pratibha College)"
        android:inputType="textPostalAddress" />

    <Button
        android:id="@+id/searchBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Search on Maps" />

</LinearLayout>

```

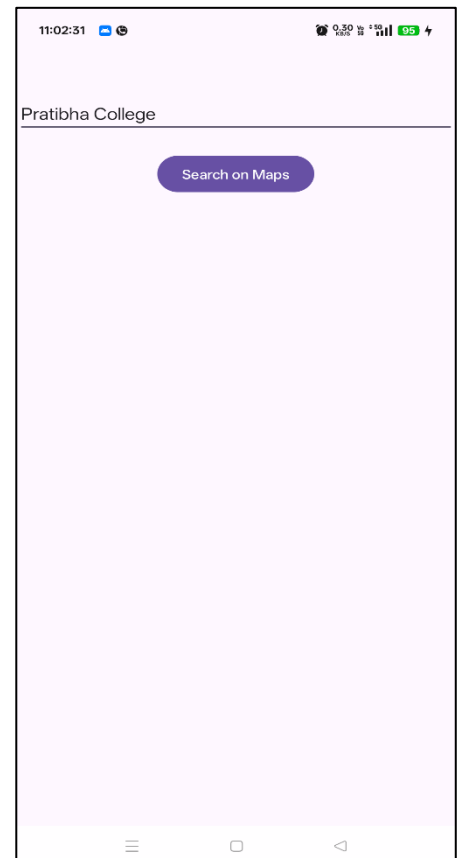
MainActivity.java

```

package com.example.gmap;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

```



```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

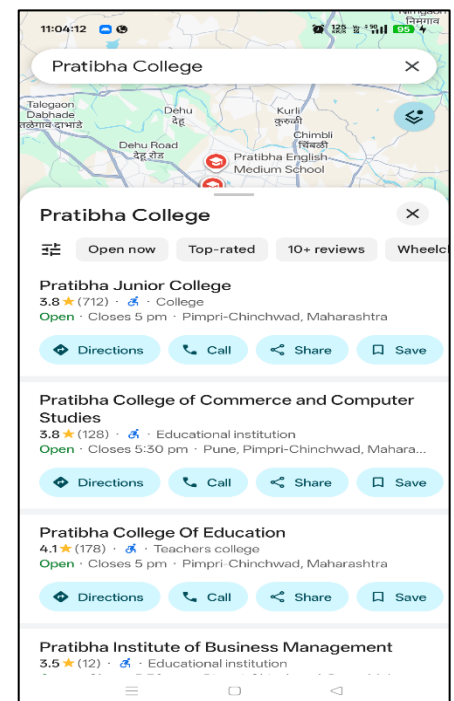
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Apply window insets for edge-to-edge support
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        EditText locationInput = findViewById(R.id.locationInput);
        Button searchBtn = findViewById(R.id.searchBtn);

        searchBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get text from input and launch Google Maps Intent
                String location = locationInput.getText().toString();
                Uri uri = Uri.parse("geo:0,0?q=" + location);
                Intent intent = new Intent(Intent.ACTION_VIEW, uri);
                startActivity(intent);
            }
        });
    }
}

```



Slip 7

Q.1) Java Android Program to Demonstrate ProgressBar.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="match_parent"

```

```

android:layout_height="25dp"
android:max="100"
android:progress="0"
android:layout_marginHorizontal="30dp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.0"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.096" />

<Button
    android:id="@+id/startBtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Start Progress"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/progressBar"
    app:layout_constraintVertical_bias="0.1" />

```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```

package com.example.progressbar;

import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.concurrent.atomic.AtomicInteger;

public class MainActivity extends AppCompatActivity {

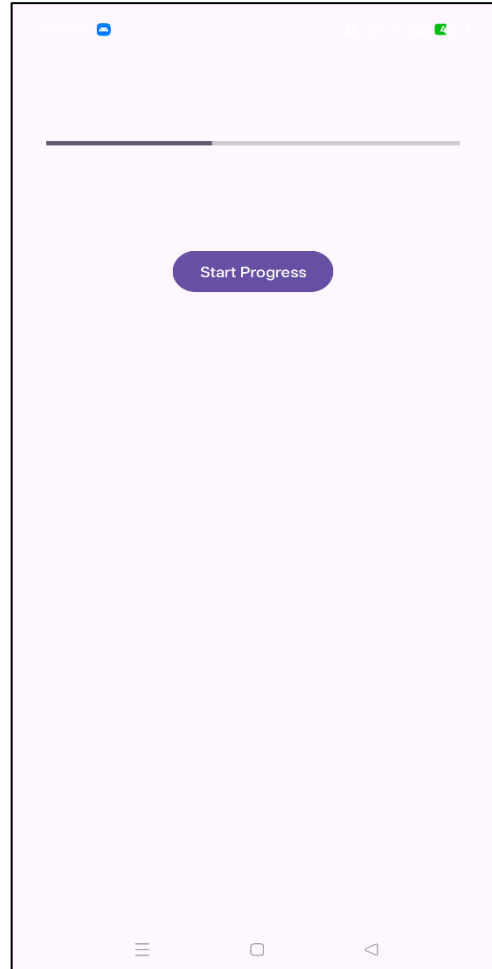
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        ProgressBar progressBar = findViewById(R.id.progressBar);
        Button startBtn = findViewById(R.id.startBtn);

        startBtn.setOnClickListener(v -> {

```




```

// Reset progress to 0 each time the button is clicked
progressBar.setProgress(0);

AtomicInteger progressStatus = new AtomicInteger();
Handler handler = new Handler();

new Thread() -> {
    while (progressStatus.get() < 100) {
        progressStatus.addAndGet(10);

        try {
            Thread.sleep(500);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

        handler.post() -> progressBar.setProgress(progressStatus.get());
    }
}).start();
});
}
}

```

Q.2] Create table Employee (E_id, name, address, ph_no). Create Application for performing the following operation on the table. (Using SQLite database). i] Insert record of 5 new Employees. ii] Show all the details of Employee.

DBHelper.java

```

package com.example.employee;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

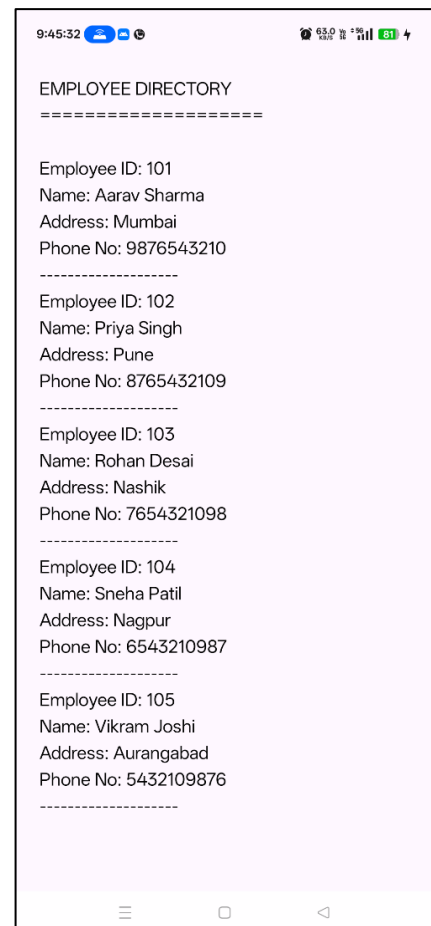
    public static final String DB_NAME = "Employee.db";
    public static final int DB_VERSION = 1;
    public static final String TABLE_NAME = "Employee";

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String query = "CREATE TABLE " + TABLE_NAME + " (" +
            "E_id INTEGER PRIMARY KEY, " +
            "name TEXT, " +
            "address TEXT, " +
            "ph_no TEXT)";
        db.execSQL(query);
    }

    @Override

```



```

public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
    onCreate(db);
}

// Method to insert a single employee
public void insertEmployee(int id, String name, String address, String ph_no) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues cv = new ContentValues();
    cv.put("E_id", id);
    cv.put("name", name);
    cv.put("address", address);
    cv.put("ph_no", ph_no);
    db.insert(TABLE_NAME, null, cv);
}

// Method to insert 5 specific records (Operation i)
public void insertFiveEmployees() {
    // Using INSERT OR IGNORE logic is best practice, but for simplicity we'll just insert
    // Note: Running this multiple times will fail on the primary key unless you clear data,
    // which is standard for this type of academic assignment.
    try {
        insertEmployee(101, "Aarav Sharma", "Mumbai", "9876543210");
        insertEmployee(102, "Priya Singh", "Pune", "8765432109");
        insertEmployee(103, "Rohan Desai", "Nashik", "7654321098");
        insertEmployee(104, "Sneha Patil", "Nagpur", "6543210987");
        insertEmployee(105, "Vikram Joshi", "Aurangabad", "5432109876");
    } catch (Exception e) {
        // Records likely already exist
    }
}

// Method to fetch all employees (Operation ii)
public Cursor getAllEmployees() {
    SQLiteDatabase db = this.getReadableDatabase();
    return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"

```

```

        android:layout_height="wrap_content"
        android:padding="24dp"
        android:textColor="#000000"
        android:textSize="18sp"
        android:lineSpacingExtra="6dp" />

```

```
</ScrollView>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.employeeedb;
```

```

import android.database.Cursor;
import android.os.Bundle;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```

```
public class MainActivity extends AppCompatActivity {
```

```

    TextView textView;
    DBHelper dbHelper;

```

```
@Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

```

```

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });

```

```

    textView = findViewById(R.id.textView);
    dbHelper = new DBHelper(this);

```

```

// Operation i: Insert 5 new employees
dbHelper.insertFiveEmployees();

```

```

// Operation ii: Show all details of Employee
Cursor cursor = dbHelper.getAllEmployees();
StringBuilder data = new StringBuilder();

```

```

data.append("EMPLOYEE DIRECTORY\n");
data.append("=====\n\n");

```

```

if (cursor.moveToFirst()) {
    do {
        data.append("Employee ID: ").append(cursor.getInt(0)).append("\n");
        data.append("Name: ").append(cursor.getString(1)).append("\n");
        data.append("Address: ").append(cursor.getString(2)).append("\n");
        data.append("Phone No: ").append(cursor.getString(3)).append("\n");
        data.append("-----\n");
    } while (cursor.moveToNext());
}

```

```

    } else {
        data.append("No employees found in the database.");
    }

    cursor.close();
    // Display the formatted data on the screen
    textView.setText(data.toString());
}
}

```

Slip 8

Q.1) Create a Application which shows Life Cycle of Activity.

MainActivity.java

```

package com.example.slip17_1;

import android.os.Bundle;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        Toast.makeText(this, "onCreate Method", Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onStart() {
        super.onStart();
        Toast.makeText(this, "onStart Method", Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(this, "onResume Method", Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(this, "onPause Method", Toast.LENGTH_SHORT).show();
    }
}

```

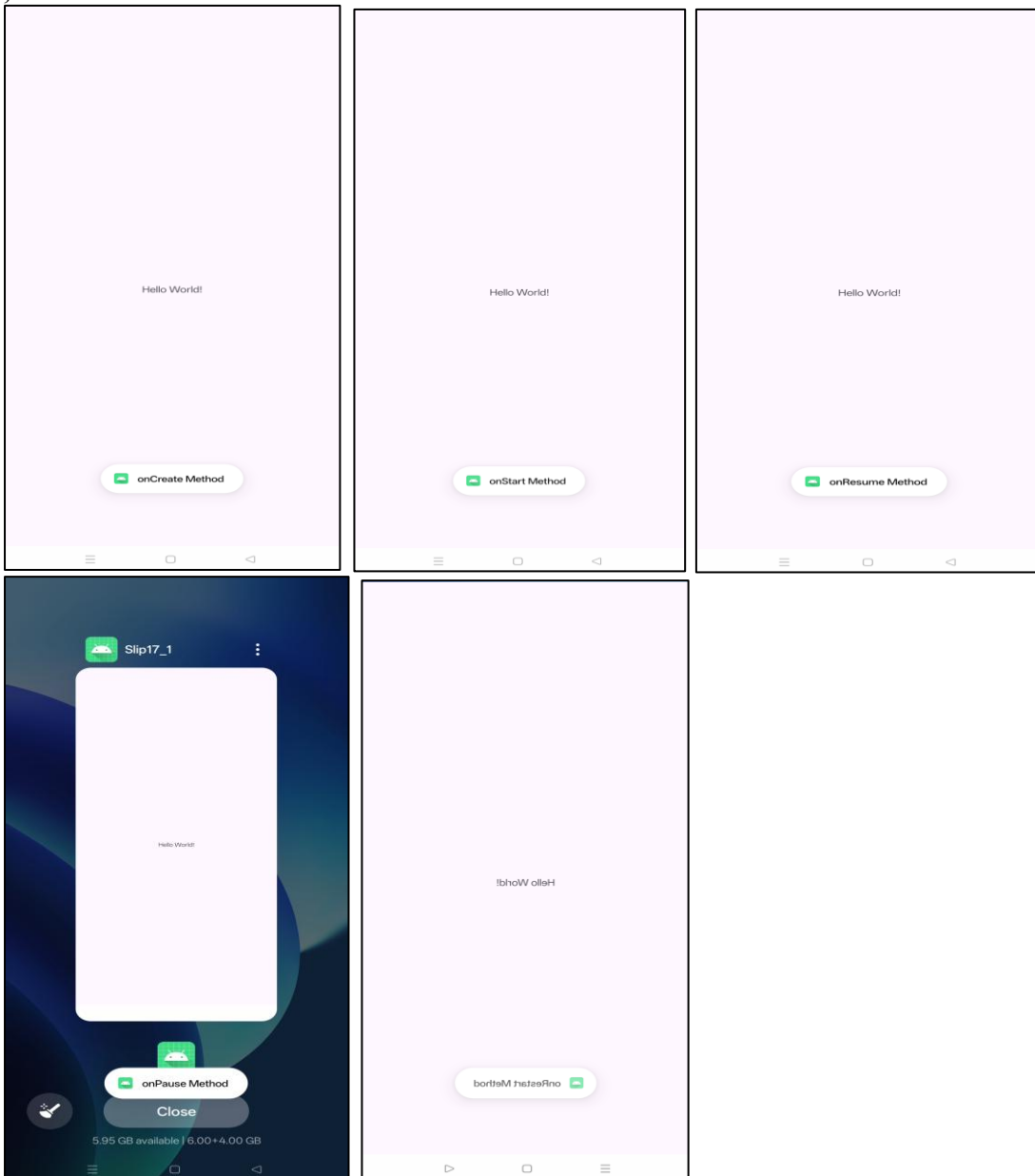
```

@Override
protected void onStop() {
    super.onStop();
    Toast.makeText(this, "onStop Method", Toast.LENGTH_SHORT).show();
}

// Added onRestart to complete the lifecycle demonstration
@Override
protected void onRestart() {
    super.onRestart();
    Toast.makeText(this, "onRestart Method", Toast.LENGTH_SHORT).show();
}

@Override
protected void onDestroy() {
    super.onDestroy();
    Toast.makeText(this, "onDestroy Method", Toast.LENGTH_SHORT).show();
}
}

```



Q.2) Create table Customer (id, name, address, ph_no). Create Application for performing the following operation on the table. (Using SQLite database). i] Insert new customer details (At least 5 records). ii] Show all the customer details.

DBHelper.java

```
package com.example.customerdb;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "Customer.db";
    public static final int DB_VERSION = 1;
    public static final String TABLE_NAME = "Customer";

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String query = "CREATE TABLE " + TABLE_NAME + " (" +
            "id INTEGER PRIMARY KEY, " +
            "name TEXT, " +
            "address TEXT, " +
            "ph_no TEXT)";
        db.execSQL(query);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    // Method to insert a single customer
    public void insertCustomer(int id, String name, String address, String ph_no) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("id", id);
        cv.put("name", name);
        cv.put("address", address);
        cv.put("ph_no", ph_no);
        db.insert(TABLE_NAME, null, cv);
    }

    // Operation i: Insert new customer details (At least 5 records)
    public void insertFiveCustomers() {
        // We use a try-catch so the app doesn't crash if you run it twice
        // and the Primary Key (id) already exists.
        try {
            insertCustomer(1, "Ramesh Patil", "Mumbai", "9876543210");
            insertCustomer(2, "Suresh Sharma", "Pune", "8765432109");
            insertCustomer(3, "Anita Desai", "Nagpur", "7654321098");
            insertCustomer(4, "Kavita Singh", "Nashik", "6543210987");
        }
    }
}
```

```

        insertCustomer(5, "Mohan Joshi", "Thane", "5432109876");
    } catch (Exception e) {
        e.printStackTrace();
    }
}

// Operation ii: Show all the customer details
public Cursor getAllCustomers() {
    SQLiteDatabase db = this.getReadableDatabase();
    return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:padding="24dp"
            android:textColor="#000000"
            android:textSize="18sp"
            android:lineSpacingExtra="6dp" />

    </ScrollView>

</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java

```

package com.example.customerdb;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    TextView textView;
    DBHelper dbHelper;

```



```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });

    textView = findViewById(R.id.textView);
    dbHelper = new DBHelper(this);

    // Operation i: Insert 5 new customers
    dbHelper.insertFiveCustomers();

    // Operation ii: Show all details of Customer
    Cursor cursor = dbHelper.getAllCustomers();
    StringBuilder data = new StringBuilder();

    data.append("CUSTOMER DIRECTORY\n");
    data.append("=====\n\n");

    if (cursor.moveToFirst()) {
        do {
            data.append("Customer ID: ").append(cursor.getInt(0)).append("\n");
            data.append("Name: ").append(cursor.getString(1)).append("\n");
            data.append("Address: ").append(cursor.getString(2)).append("\n");
            data.append("Phone No: ").append(cursor.getString(3)).append("\n");
            data.append("-----\n");
        } while (cursor.moveToNext());
    } else {
        data.append("No customers found in the database.");
    }

    cursor.close();

    // Display the formatted data on the screen
    textView.setText(data.toString());
}
}

```

Slip 9

Q.1) Create an application that allows the user to enter a number in the textbox named „getnum“. Check whether the number in the textbox „getnum“ is Palindrome or not. Print the message accordingly in the label when the user clicks on the button „Check“.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

```



```

android:gravity="center"
android:orientation="vertical"
android:padding="24dp">

```

```

<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Enter a Number:"
    android:textSize="18sp"
    android:textStyle="bold" />

```

```

<EditText
    android:id="@+id/getnum"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="e.g. 12321"
    android:inputType="number"
    android:textAlignment="center"
    android:textSize="18sp" />

```

```

<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Check Palindrome"
    android:textSize="16sp"
    android:textStyle="bold" />

```

```

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="30dp"
    android:text=""
    android:textColor="#000000"
    android:textSize="20sp"
    android:textStyle="bold" />

```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.palindrome_toast;
```

```

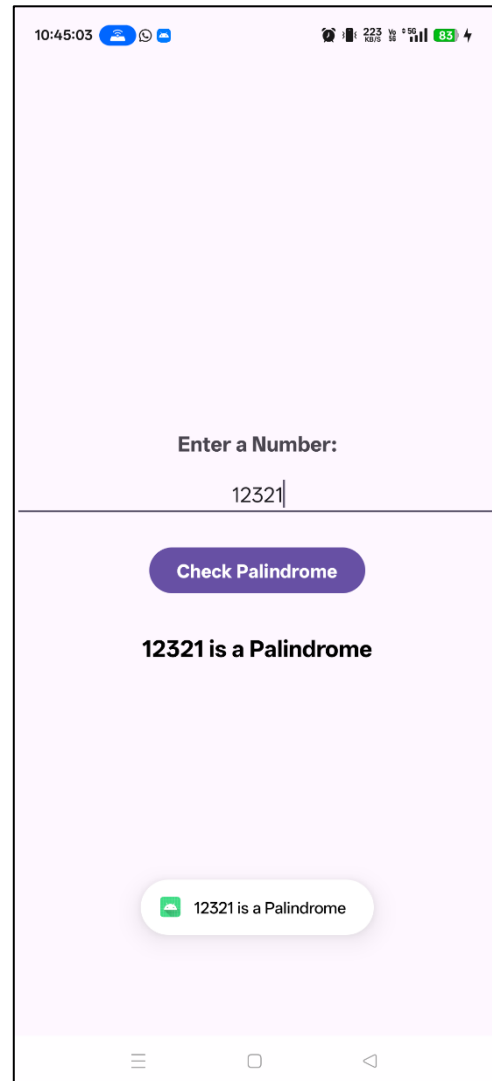
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

```

```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```



```

public class MainActivity extends AppCompatActivity {

    EditText e1;
    Button b1;
    TextView t1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        // Updated ID to match your requirement: getnum
        e1 = findViewById(R.id.getnum);
        b1 = findViewById(R.id.button);
        t1 = findViewById(R.id.textView2);

        b1.setOnClickListener(new View.OnClickListener() {
            @SuppressWarnings("SetTextI18n")
            @Override
            public void onClick(View view) {

                String input = e1.getText().toString();

                // Prevent crash if the input is empty
                if (input.isEmpty()) {
                    t1.setText("Please enter a number");
                    return;
                }

                boolean isPalindrome = true;
                int start = 0;
                int end = input.length() - 1;

                // Two-pointer approach to check for palindrome
                while (start <= end) {
                    if (input.charAt(start) != input.charAt(end)) {
                        isPalindrome = false;
                        break;
                    }
                    start++;
                    end--;
                }

                // Display results on screen and via Toast
                if (isPalindrome) {
                    t1.setText(input + " is a Palindrome");
                    Toast.makeText(MainActivity.this, input + " is a Palindrome", Toast.LENGTH_LONG).show();
                } else {
                    t1.setText(input + " is NOT a Palindrome");
                    Toast.makeText(MainActivity.this, input + " is NOT a Palindrome", Toast.LENGTH_LONG).show();
                }
            }
        });
    }
}

```

```
}
}
```

Q.2) Java Android Program to perform all arithmetic Operations using Calculator.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="0"
        android:inputType="numberDecimal|numberSigned"
        android:textSize="24sp" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="TextView"
        android:textSize="20sp"
        android:layout_marginBottom="30dp" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginBottom="10dp">

        <Button
            android:id="@+id/btnAdd"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:layout_marginEnd="10dp"
            android:text="+" />

        <Button
            android:id="@+id/btnSub"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:layout_marginEnd="100dp"
            android:text="-" />

    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginBottom="10dp">

        <Button
            android:id="@+id/btnDiv"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:layout_marginEnd="10dp"
            android:text="/" />

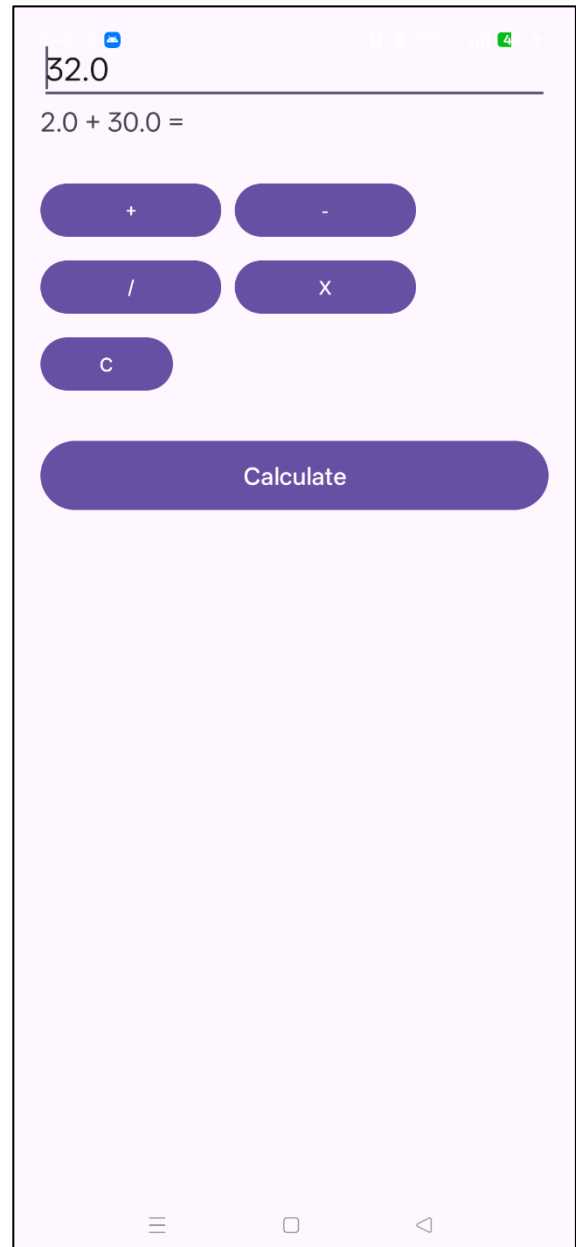
        <Button
            android:id="@+id/btnMul"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:layout_marginEnd="10dp"
            android:text="x" />

        <Button
            android:id="@+id/btnC"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:layout_marginEnd="10dp"
            android:text="C" />

        <Button
            android:id="@+id/btnCalc"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:layout_marginEnd="10dp"
            android:text="Calculate" />

    </LinearLayout>

</LinearLayout>
```



```
    android:layout_width="0dp"
    android:layout_weight="1"
    android:layout_height="wrap_content"
    android:layout_marginEnd="10dp"
    android:text="/" />
```

```
<Button
    android:id="@+id/btnMul"
    android:layout_width="0dp"
    android:layout_weight="1"
    android:layout_height="wrap_content"
    android:layout_marginEnd="100dp"
    android:text="X" />
</LinearLayout>
```

```
<Button
    android:id="@+id/btnClear"
    android:layout_width="100dp"
    android:layout_height="wrap_content"
    android:layout_marginBottom="30dp"
    android:text="C" />
```

```
<Button
    android:id="@+id/btnCalc"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:text="Calculate"
    android:textSize="18sp" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.calculator2;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etInput;
    TextView tvResult;
    Button btnAdd, btnSub, btnMul, btnDiv, btnClear, btnCalc;

    double num1 = Double.NaN;
    double num2;
    String operator = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI components
        etInput = findViewById(R.id.etInput);
```

```

tvResult = findViewById(R.id.tvResult);
btnAdd = findViewById(R.id.btnAdd);
btnSub = findViewById(R.id.btnSub);
btnMul = findViewById(R.id.btnMul);
btnDiv = findViewById(R.id.btnDiv);
btnClear = findViewById(R.id.btnClear);
btnCalc = findViewById(R.id.btnCalc);

// Unified click listener for all operator buttons
View.OnClickListener operatorListener = new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (etInput.getText().toString().isEmpty()) return;

        // Save the first number and the chosen operator
        num1 = Double.parseDouble(etInput.getText().toString());
        Button b = (Button) v;
        operator = b.getText().toString();

        // Show the current status in the TextView and clear the EditText
        tvResult.setText(num1 + " " + operator);
        etInput.setText("");
    }
};

// Assign the listener to the operator buttons
btnAdd.setOnClickListener(operatorListener);
btnSub.setOnClickListener(operatorListener);
btnMul.setOnClickListener(operatorListener);
btnDiv.setOnClickListener(operatorListener);

// Clear Button Logic
btnClear.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        num1 = Double.NaN;
        num2 = 0;
        operator = "";
        etInput.setText("");
        tvResult.setText("TextView");
    }
});

// Calculate Button Logic
btnCalc.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        // Ensure the user has entered a first number, an operator, and a second number
        if (Double.isNaN(num1) || etInput.getText().toString().isEmpty()) {
            Toast.makeText(MainActivity.this, "Please enter an equation", Toast.LENGTH_SHORT).show();
            return;
        }

        num2 = Double.parseDouble(etInput.getText().toString());
        double result = 0;

        // Perform the math based on the saved operator
        switch (operator) {
            case "+":
                result = num1 + num2;

```

```

        break;
    case "-":
        result = num1 - num2;
        break;
    case "X":
        result = num1 * num2;
        break;
    case "/":
        if (num2 == 0) {
            Toast.makeText(MainActivity.this, "Cannot divide by zero!", Toast.LENGTH_SHORT).show();
            return;
        }
        result = num1 / num2;
        break;
    }

    // Display the final equation in the TextView and the result in the EditText
    tvResult.setText(num1 + " " + operator + " " + num2 + "=");
    etInput.setText(String.valueOf(result));

    // Allow the user to chain the next calculation using this result
    num1 = result;
    operator = "";
}
});
}
}

```

Slip 10

Q.1] Create an application that allows the user to enter a number in the textbox named getnum. Check whether the number in the textbox getnum is Armstrong or not. Print the message using Toast control when the user clicks on the button Check.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Enter a Number:"
        android:textSize="18sp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/getnum"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:hint="e.g. 153"
    >

```

```

android:inputType="number"
android:textAlignment="center"
android:textSize="18sp" />

```

```

<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Check Armstrong"
    android:textSize="16sp"
    android:textStyle="bold" />

```

```

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="30dp"
    android:text=""
    android:textColor="#000000"
    android:textSize="20sp"
    android:textStyle="bold" />

```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.armstrong_toast;
```

```

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

```

```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```

```
public class MainActivity extends AppCompatActivity {
```

```

    EditText e1;
    Button b1;
    TextView t1;

```

```
@Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

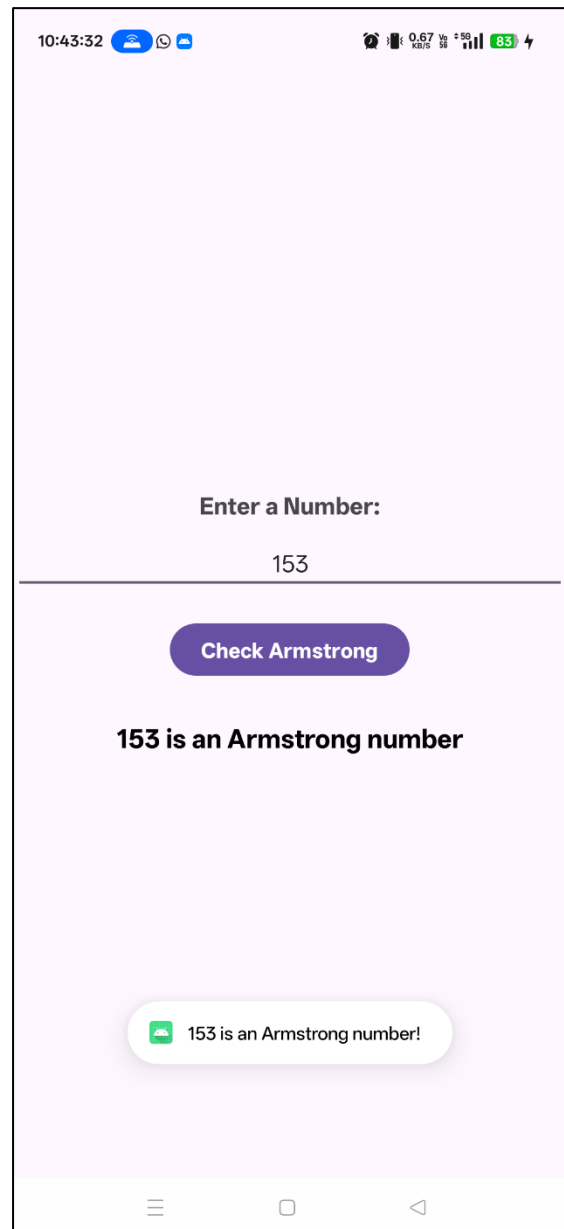
```

```

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });

```

```
// Linked to the specific ID requested: getnum
```



```

e1 = findViewById(R.id.getnum);
b1 = findViewById(R.id.button);
t1 = findViewById(R.id.textView2);

b1.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        String input = e1.getText().toString();

        // Safety check: Prevent app crash if input is empty
        if (input.isEmpty()) {
            Toast.makeText(MainActivity.this, "Please enter a number first", Toast.LENGTH_SHORT).show();
            return;
        }

        int num = Integer.parseInt(input);
        int temp = num;
        int digits = 0;
        int result = 0;

        // Step 1: Count the number of digits
        while (temp != 0) {
            digits++;
            temp /= 10;
        }

        temp = num; // Reset temp back to the original number

        // Step 2: Calculate Armstrong sum
        while (temp != 0) {
            int remainder = temp % 10;
            result += (int) Math.pow(remainder, digits);
            temp /= 10;
        }

        // Step 3: Check and display result using Toast (and updating the TextView)
        if (result == num) {
            t1.setText(num + " is an Armstrong number");
            Toast.makeText(MainActivity.this, num + " is an Armstrong number!", Toast.LENGTH_LONG).show();
        } else {
            t1.setText(num + " is NOT an Armstrong number");
            Toast.makeText(MainActivity.this, num + " is NOT an Armstrong number",
Toast.LENGTH_LONG).show();
        }
    }
});
}
}

```

Q.2] Write a program to draw GUI by using Spinner, Buttons.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

```



```

<EditText
    android:id="@+id/inputText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter a new item"
    android:layout_marginBottom="16dp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center"
    android:layout_marginBottom="24dp">

    <Button
        android:id="@+id/addButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"
        android:layout_marginEnd="16dp"/>

    <Button
        android:id="@+id/removeButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Remove" />
</LinearLayout>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Select Item:"
    android:layout_marginBottom="8dp"/>

<Spinner
    android:id="@+id/spinner"
    android:layout_width="match_parent"
    android:layout_height="48dp"
    android:background="@android:drawable/btn_dropdown" />

</LinearLayout>

```



MainActivity.java

```

package com.example.dynamicspinner;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.ArrayList;

```

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Handle window insets for edge-to-edge display
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        EditText inputText = findViewById(R.id.inputText);
        Button addButton = findViewById(R.id.addButton);
        Button removeButton = findViewById(R.id.removeButton);
        Spinner spinner = findViewById(R.id.spinner);

        // Set up the ArrayList and Adapter for the Spinner
        ArrayList<String> itemsList = new ArrayList<>();
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this, android.R.layout.simple_spinner_item, itemsList);
        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
        spinner.setAdapter(adapter);

        // Add Button Click Listener
        addButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String newItem = inputText.getText().toString();
                if (!newItem.isEmpty()) {
                    itemsList.add(newItem);
                    adapter.notifyDataSetChanged(); // Refreshes the spinner to show the new item
                    inputText.setText(""); // Optional: clears the input field after adding
                }
            }
        });

        // Remove Button Click Listener
        removeButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int position = spinner.getSelectedItemPosition();
                // Check if the spinner actually has a valid item selected
                if (position != Spinner.INVALID_POSITION) {
                    itemsList.remove(position);
                    adapter.notifyDataSetChanged(); // Refreshes the spinner after deletion
                }
            }
        });
    }
}

```